



How organisational and team climates foster design thinking for project success

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ABSTRACT

Traditional project management methods struggle with the uncertainties and complexities of modern, innovative projects. Research highlights the need for flexible approaches integrating exploration and learning, such as design thinking. However, implementing design thinking practices in project management faces various obstacles. Beyond structural and strategic hurdles, the organisational context—specifically how individuals experience the work environment—likely plays a critical role in shaping the intensity and success of such practices. This study analyses the organisational and team climates that support implementing design thinking practices to increase project success while considering the role of project contingencies. Drawing on data from 379 projects nested in 93 companies, we investigate the impact of organisational climates (supportive climate, innovation climate, open climate) and team climates (servant leadership, team autonomy). Our findings suggest that design thinking practices improve project success. At the same time, we identify a supportive climate, an open climate, servant leadership, and team autonomy as conducive to design thinking practices. In addition, design thinking practices are more beneficial for project success when project innovativeness is high. Project innovativeness also strengthens the effect of a supportive climate on design thinking practices. We contribute to the literature on project management and design thinking by providing the first quantitative evidence on how specific climate conditions within the broader organisational context support the application of design thinking practices in project environments.

1. Introduction

As innovation and technological shifts reshape project management, traditional methods—often reliant on explicitly delineated objectives—struggle to cope with the uncertainty and complexity of modern projects. These challenges are especially pronounced in contexts involving emergent technologies, novel domains, and ambiguous customer specifications (Atkinson et al., 2006; BenMahmoud-Jouini & Carlgren, 2023). Scholars have thus called for approaches that balance structured delivery with exploration and learning (Lenfle, 2008). In response, design thinking has emerged as a promising approach, offering iterative, user-centred strategies for tackling ambiguity and complexity in project settings (Brown, 2008; Dorst, 2011).

However, despite its promise, implementing design thinking practices in projects is far from straightforward. Research shows that organisations face significant challenges when adopting design thinking, and not all projects achieve success (Ben Mahmoud-Jouini et al., 2016; Hölzle & Rhinow, 2019; Przybilla et al., 2020). Its implementation often

clashes with formal project management systems, particularly when iterative processes are difficult to align with predefined milestones (Ben Mahmoud-Jouini et al., 2016; Hölzle & Rhinow, 2019). Moreover, design thinking can be resource-intensive, demanding significant time and cost investments (Carlgren, Elmquist, et al., 2016a; Mahmoud-Jouini et al., 2019; Rauth et al., 2015). These challenges are particularly pronounced in large organisations, where structural, strategic, and cultural constraints can undermine design thinking's success (Carlgren, Elmquist, et al., 2016a; Dell'Era et al., 2020; Dunne, 2018; Hölzle & Rhinow, 2019). Organisations frequently perceive design thinking as merely a set of tools, which overlooks the broader organisational conditions required to support its use (Ansari et al., 2010; Canato et al., 2013; Carlgren & BenMahmoud-Jouini, 2022).

Although scholars have begun to explore which organisational cultures may be more conducive to design thinking's application (Carlgren & BenMahmoud-Jouini, 2022; Elsbach & Stigliani, 2018; Wrigley et al., 2020), much of this work remains anecdotal (Carlgren & BenMahmoud-Jouini, 2022; Elsbach & Stigliani, 2018; Wrigley et al., 2020) and

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generalisable evidence is still lacking. This is partly due to the difficulty of measuring design thinking, given that its success is contingent on subjective judgement and qualitative factors, such as creativity and user satisfaction, which are difficult to quantify.

This study aims to fill this gap by providing quantitative, empirical insights into the organisational and team climates that support the implementation of design thinking in project environments. We build on previous work (Roth et al., 2020; Salzmann & Kock, 2018) to operationalise and measure design thinking practices at the project level. More specifically, we focus on how individuals experience their work environment and how this influences the adoption of design thinking practices. Our first research question is: *Which organisational and team climates support the implementation of design thinking practices in projects?*

In addition, we explore the role of project contingencies. While numerous studies highlight the benefits of design thinking, there is an ongoing debate about whether its practices are better suited for radical or incremental innovation. On the one hand, design thinking may particularly catalyse highly innovative projects by creatively exploring constraints and tackling complex, uncertain challenges (Ben Mahmoud-Jouini et al., 2016; Brown, 2008; Gruber et al., 2015). On the other hand, design thinking could be suited only for less innovative, incremental projects, as its iterative and user-oriented approach inherently fosters gradual improvements by building on prior ideas (Norman & Verganti, 2014; Verganti, 2011). To address this debate, we ask a second research question: *How does project innovativeness influence the relationship between climates and design thinking, and between design thinking and project success?*

The study makes several contributions. First, the study enhances project management theory by highlighting the limitations of traditional approaches in handling innovative, uncertain, and exploratory projects (Atkinson et al., 2006; Ben Mahmoud-Jouini & Carlgren, 2023; Lenfle, 2008). This is particularly relevant as organisations increasingly face dynamic and complex challenges that demand more adaptive methodologies (Atkinson et al., 2006; Ben Mahmoud-Jouini & Carlgren, 2023; Lenfle, 2008). Second, by providing large-scale empirical insights, the study adds clarity to the ongoing debate on whether design thinking is better suited for radical or incremental innovation (Micheli et al., 2019; Norman & Verganti, 2014; Verganti, 2011). Third, it deepens our understanding of the organisational barriers to design thinking (Ben Mahmoud-Jouini et al., 2016; Hölzle & Rhinow, 2019) by identifying organisational and team climates that help overcome them. Finally, the research offers practical value: organisations can use our findings to evaluate whether their current environments are conducive to design thinking and what changes may be needed to support it better.

2. Conceptual background

2.1. Role of uncertainties and exploration in project management

Project managers face the ongoing challenge of managing uncertainties, vagueness and ambiguity (Atkinson et al., 2006; Chapman & Ward, 2003). These challenges can arise from various factors, such as unclear project specifications and requirements, project novelty, lack of experience, and project complexity (Armor & Taylor, 2002; Atkinson et al., 2006; Buehler et al., 2002). Each project lifecycle phase presents distinct uncertainties, risks and challenges (Atkinson et al., 2006; Chapman & Ward, 2003). The literature contends that traditional project management practices often fail to adequately address the uncertainties inherent across all project lifecycle phases (Atkinson et al., 2006; Chapman & Ward, 2003). While traditional project management approaches rely on clearly defined goals and timelines, they frequently fall short when applied to novel and innovative projects (Lenfle, 2008).

Additionally, project objectives and requirements often evolve throughout the project lifecycle or may remain unclear even as the project progresses, contrary to initial assumptions. Atkinson (2006) highlights that "soft projects"—involving human interactions, social

dynamics, and evolving objectives—are particularly prone to heightened uncertainty and ambiguity. Meanwhile, Lenfle (2008) work focuses on exploration projects involving new and innovative technical developments, practices, and business models, all involving significant uncertainty and instability. In such novel, innovative, and exploratory projects, customer requirements and technologies are often unknown, underscoring the importance of flexible and adaptive managerial project management for success (Atkinson et al., 2006; Lenfle, 2008). Further, Atkinson (2006) and Lenfle (2008) describe the vital role that organisational values play in managing uncertainty. They note when an organisation fosters openness, adaptability, and a commitment to learning, it becomes better equipped to handle uncertainty effectively.

In summary, previous literature shows that traditional project management reaches its limits when dealing with uncertainty in innovative, novel, and exploratory projects. Such projects require flexible and adaptive approaches that support an environment of openness to failure and uncertainty.

2.2. Role of design thinking in project management

With its strong emphasis on user orientation and exploration, design thinking emerges as a promising problem-solving approach for addressing uncertainties and ambiguity, ultimately improving project success (Ben Mahmoud-Jouini et al., 2016; Ben Mahmoud-Jouini & Carlgren, 2023; Liedtka & Locatelli, 2023). The project management literature indicates that design thinking practices may be particularly beneficial for complex projects (Ben Mahmoud-Jouini & Carlgren, 2023; Liedtka & Locatelli, 2023). They can facilitate project planning, delivery and stakeholder management, particularly in projects involving several stakeholders and opinions, as is often the case in complex projects (Liedtka & Locatelli, 2023). The human-centric approach, the importance of diverse perspectives, and the utilisation of experimentation and visualisation practices driven by design thinking are all valuable tools in complex projects (Ben Mahmoud-Jouini & Carlgren, 2023).

Previous literature describes the potential benefits and dilemmas of introducing design thinking to project management and indicates that not all design thinking projects lead to success (Ben Mahmoud-Jouini et al., 2016; Hölzle & Rhinow, 2019; Przybilla et al., 2020). With its underlying tools and mindset, design thinking fosters the skills to balance the values of consistency and relevance and thus strengthens organisational ambidexterity (Beverland et al., 2015; Randhawa et al., 2021). However, design thinking encounters various obstacles regarding structural and strategic project management guidelines. The iterative way of working is sometimes difficult to reconcile with project management milestones (Ben Mahmoud-Jouini et al., 2016; Hölzle & Rhinow, 2019), and its implementation can be highly cost- and time-consuming (Carlgren, Elmquist, et al., 2016b; Mahmoud-Jouini et al., 2019; Rauth et al., 2015). Despite these challenges, several studies suggest that it is worthwhile for organisations to invest in integrating design thinking, particularly in the long term, to foster project success and innovation outcomes (Ben Mahmoud-Jouini et al., 2016; Ben Mahmoud-Jouini & Carlgren, 2023; Carlgren, Elmquist, et al., 2016a; Johansson-Sköldberg et al., 2013; Liedtka & Locatelli, 2023).

2.3. Definition of design thinking practices

Design thinking is a creative and innovative work approach driven by exploration and user-centricity (Brown, 2008; Magistretti et al., 2021). The concept draws from the designers' way of thinking and working and is understood as a set of tools, practices, and mindsets (Brown & Katz, 2011; Johansson-Sköldberg et al., 2013; Micheli et al., 2019). The literature describes three fundamental categories of design thinking (Brown, 2008; Seidel & Fixson, 2013): (1) need-finding tools (e.g., intensive customer research and interviews, customer journey mappings), (2) idea generation tools (e.g., brainstorming and co-creation sessions to reframe the initial problem), and (3) idea-testing tools (e.

g., experimentation, prototyping, and testing of the developed ideas and products). Design thinking involves a shift between divergent and convergent thinking (Brown & Katz, 2011; Liedtka, 2018), making it particularly effective in addressing ill-defined or wicked problems (Buchanan, 1992) that are ambiguous with multiple possible solutions (Goldschmidt, 1997).

In this study, we focus on the five design thinking themes from Carlgren et al. (2016): user focus, problem framing, visualisation, experimentation, and diversity. The concept of *user focus* entails cultivating empathy for users to discern latent needs and prioritise them within the developmental process (Beverland et al., 2015; Dalton & Kahute, 2016; Veryzer & Borja de Mozota, 2005). Extensive user research, observations, interviews, and interactions with relevant parties are instrumental in acquiring these in-depth insights (Arnould & Cayla, 2015; Carlgren, Rauth, et al., 2016; Goffin et al., 2012). *Problem framing* is a process that encourages the interrogation and re-framing of initial concepts. Problem framing aims to shed light on the original problem from diverse vantage points and broaden the initial problem's boundaries (Carlgren, Elmquist, et al., 2016a; Liedtka, 2015). Employing "what if?" inquiries can be instrumental in fostering these holistic perspectives (Carlgren, Rauth, et al., 2016). *Visualisation* means making abstract ideas visual and tangible at the onset of the development process (Carlgren, Rauth, et al., 2016). These visualisations are typically rudimentary, often taking the form of images, sketches, or LEGO prototypes, and serve merely to enhance the comprehensibility of the concept (Brown, 2008; Carlgren, Rauth, et al., 2016; Dalton & Kahute, 2016). The iterative and flexible nature of *experimentation*, marked by early prototyping and user testing, is instrumental in facilitating a dynamic and adaptable development process (Brown, 2008; Carlgren, Rauth, et al., 2016). This approach encourages a mindset that embraces frequent and early experimentation, often characterised by a playful and optimistic disposition when confronted with setbacks (Carlgren, Rauth, et al., 2016; Dalton & Kahute, 2016; Rylander, 2009). *Diversity* is the extent to which team members are drawn from various academic and professional backgrounds, including educational attainment, knowledge levels, and social categories such as nationality or personality (Brown, 2008; Carlgren, Rauth, et al., 2016). The framing proposed by Carlgren et al. (2016) is particularly useful for research purposes, as it attempts to capture the essential features of the work approach and make it measurable.

2.4. Design thinking and organisational climates

Canatone et al. (2013, p. 1749) identify a "recursive relationship between organisational practices—understood as patterns of day-to-day activities (what people do)—and culture—understood as a set of beliefs that people use to make and give a sense of what they do." This recursive relationship also accounts for design thinking and organisational culture, a construct of structures and processes, beliefs, values, and underlying assumptions shared by organisational members (Detert et al., 2000; Schein, 2010). Organisational climate, on the other hand, refers to the shared perceptions and meanings that employees associate with their work environment (Schneider et al., 2013). It encompasses employees' experiences and feelings about their organisation (Schneider et al., 2013). Although both constructs are related and essential for understanding organisational dynamics, organisational climate refers more to current perceptions of the work environment. In contrast, organisational culture encompasses the deeper, ingrained values and beliefs that shape an organisation over time (Schneider et al., 2013). Denison (1996) differentiates organisational culture and climate by emphasising their temporal focus. Organisational culture is rooted in historical evolution, reflecting values and beliefs that develop over time, whereas organisational climate represents a snapshot of present conditions and their immediate effects (Denison et al., 1996). As culture encompasses the underlying values and assumptions that define an organisation, it is primarily studied using qualitative methods (Denison

et al., 1996). In contrast, climate pertains to observable, surface-level manifestations of the organisational environment, making it more suitable for quantitative analysis to identify generalisable patterns (Denison et al., 1996). Organisational culture, as a foundational system of values and norms, shapes how employees perceive their work environment. These perceptions, in turn, form the basis of organisational climate, which can be empirically assessed by examining employees' views on specific workplace characteristics. While culture is a complex and abstract construct, climate is a measurable and tangible representation of its influence on organisational outcomes. Accordingly, this study seeks to capture the essence of organisational culture by analysing various dimensions of climate.

Elsbach and Stigliani (2018) analyse which norms and values support or hinder the implementation of design thinking practices in organisations. Cultures characterised by experimental and collaborative climates support the application of design thinking, while more "traditional" cultures characterised by perfectionism, productivity, and isolated specialisation tend to hinder it (Elsbach & Stigliani, 2018). The main challenge for organisations using design thinking is usually not learning how to use the tools and techniques but understanding and internalising the associated mindset. Organisations that have adopted a culture of experimentation and collaboration not only support the application of design thinking, but their underlying culture and mindset lead more or less naturally to its application (Bicen & Johnson, 2015). Ansari et al. (2010, p. 78) refer to this as "cultural fit", which is "the degree to which the characteristics of a diffusing practice are compatible with the cultural values, beliefs, and practices of potential adopters." Consequently, the efficacy of implementing design thinking in organisations is contingent on its fit with organizational norms and values.

Detert et al. (2000) analysed the interplay between total quality management and conducive organisational culture and developed an eight-dimensional framework. Building on this, Carlgren and BenMahmoud-Jouini (2022) identified a cultural archetype defining a cultural orientation that is beneficial for successfully implementing design thinking. Based on a case study, they describe the following dimensions: subjective and aesthetic ways of knowing, long-term and non-linear views about time, intrinsic motivation and a sense of purpose, flexibility and change, relationships, empathy, and emotions at work, collaboration and inclusion, team autonomy and informality, and external orientation. Further, they describe challenges and consequences related to a cultural mismatch, such as poor team climate, lack of legitimacy, and disappointing results. The case study of Wrigley et al. (2020) identifies four organisational conditions to implement design thinking successfully. They describe that strategic vision, facilities, cultural capital, and organisational policies that foster employee engagement, creativity, and collaboration are conducive to design thinking.

Summing up, several studies investigate the interplay between design thinking and organisational culture. While conceptual (Elsbach & Stigliani, 2018; Wrigley et al., 2020) and qualitative studies (Carlgren & BenMahmoud-Jouini, 2022) give profound insights on which organisational culture is conducive to design thinking practices, the literature lacks large-scale quantitative evidence. The quantification of design thinking is challenging due to its reliance on subjective factors, such as creativity and user satisfaction, which are difficult to measure. While anecdotal evidence is context-specific, quantitative methods provide consistent, replicable, and more widely applicable results. Building on earlier research (Roth et al., 2020; Salzmann & Kock, 2018), we use the five design thinking themes from Carlgren et al. (2016) to capture the essential features of the approach, making it both measurable and quantifiable. Similarly, we conceptualise organisation and team climates as observable manifestations of the prevailing organisational context (Denison et al., 1996). Using this approach, our study aims to provide large-scale empirical evidence on the influence of different climates on design thinking adoption and, eventually, project success.

3. Hypotheses development

Fig. 1 shows the study's conceptual model. Based on relevant literature (Carlgren & BenMahmoud-Jouini, 2022; Wrigley et al., 2020), we analyse the influence of organisational and team climates on design thinking practices. Further, we investigate the moderating effects of project innovativeness on the relationships between design thinking practices' antecedents and consequences.

3.1. Design thinking practices and project success

We apply the five themes from Carlgren et al. (2016) to conceptualise design thinking: user focus, problem framing, visualisation, experimentation, and diversity. This framing is especially useful for research because it tries to capture the approach's essential characteristics independent of the specific methodology. We argue for a positive relationship between design thinking practices and the overall project success, which we conceptualise by how different project goals such as profitability, learning, budget or defined specifications were achieved (Hoegl & Gemuenden, 2001).

Design thinking as an innovative problem-solving approach shows characteristics enhancing product and innovation outcomes (Brown, 2008; Nakata & Hwang, 2020). The iterative work approach with early prototyping and regular testing helps identify errors early in the development process, counteract potential development costs, and ultimately improve the product outcome. The strong user focus implied by design thinking practices encourages regular customer feedback to determine whether the product meets customer needs and how customers rate the product compared to the competition (Nakata & Hwang, 2020). Thus, design thinking practices encourage product development that meets customer and market needs, increasing customer satisfaction and sales volume. Combining convergent and divergent thinking encourages companies to regularly challenge their ideas to achieve better results (Brown & Katz, 2011; Liedtka, 2018). Team members with different backgrounds enrich the product's creativity and innovation outcomes. Combining creative thinking and an iterative, user-centric work approach strengthens product and project performance (Nakata & Hwang, 2020; Roth et al., 2020; Seidel & Fixson, 2013). Therefore, we hypothesise:

H1: Design thinking practices strengthen project success.

3.2. Organisational climates

This study analyses which organisational climates are conducive to design thinking practices. Based on relevant literature (Carlgren & BenMahmoud-Jouini, 2022), we examine the role of supportive, innovative, and open climates.

A supportive climate refers to the extent to which organisations value the employees' work and efforts and care about their well-being (Eisenberger et al., 1986; Tansky & Cohen, 2001). Such a climate fosters a positive and supportive work environment where employees are assigned tasks according to their interests and abilities (Eisenberger et al., 1986). This alignment makes employees feel that their work is meaningful and appreciated, enhancing their intrinsic motivation and commitment (Eisenberger et al., 1986; Rhoades & Eisenberger, 2002). We assume that a supportive climate fosters the implementation of design thinking practices.

Implementing a new work approach can be complex and time-consuming for employees, so higher employee motivation and engagement are conducive to its application (Carlgren & BenMahmoud-Jouini, 2022). A supportive climate not only increases employee motivation and commitment but also their level of trust and loyalty to the organisation and its decisions (Rhoades & Eisenberger, 2002). As a result, employees are more likely to recognise the value and purpose of the design thinking approach and see its implementation as an opportunity to learn. Therefore, we assume that a supportive climate fosters employees' openness and willingness to engage with the new work approach and hypothesise:

H2a: A supportive climate is conducive to using design thinking practices.

An innovative climate refers to organisations encouraging innovative and creative tasks while unconventional or controversial opinions are welcomed (Amabile, 1983; Kock et al., 2015; Kock & Georg Gemuenden, 2016). This openness and encouragement help create a more emphatic and tolerant work atmosphere in which employees feel psychologically safe (Baer & Frese, 2003; Edmondson et al., 2004). We argue that organisations with an innovative climate are more likely to adopt design thinking practices.

An innovative work environment encourages employees to take risks in generating creative and innovative ideas without fear of making mistakes. It also provides a foundation for constructive feedback and allows a certain playfulness in collaborating (Carlgren & BenMahmoud-Jouini, 2022). This work environment aligns with the associated techniques and mindset of design thinking, such as experimentation,

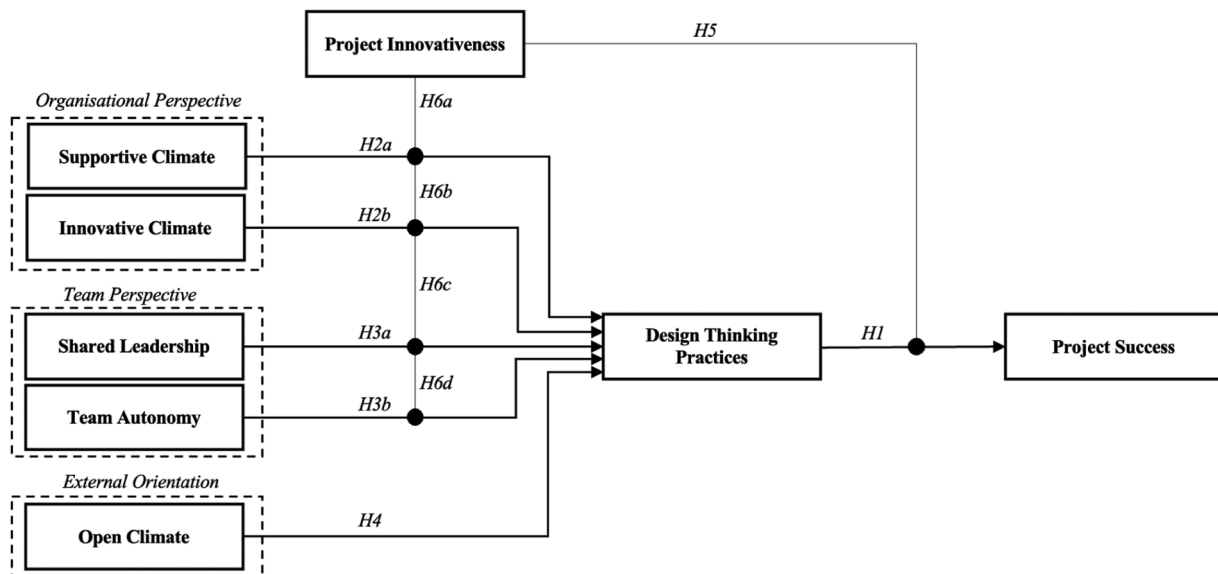


Fig. 1. Conceptual model.

openness to failure, and playfulness (Bason & Austin, 2019; Elsbach & Stigliani, 2018). The design thinking literature (Carlgrén & BenMahmoud-Jouini, 2022; Elsbach & Stigliani, 2018) indicates that organisations shaped by such innovative and open-minded values apply design thinking practices more effectively. Thus, we assume:

H2b: An innovative climate is conducive to using design thinking practices.

Design thinking practices strongly rely on the organisation's environment and external sources, such as users, stakeholders, and competitors. Drawing on relevant literature (Carlgrén & BenMahmoud-Jouini, 2022), we explore the role of an open climate in fostering design thinking practices. An open climate refers to an organisational mindset that embraces collaboration with external partners and actively seeks ideas and technologies beyond its organisational boundaries (Chesbrough, 2003; de Araújo Burcharth et al., 2014; Laursen & Salter, 2014). Such organisations assess innovation and success not solely through internal perspectives, such as those of managers or engineers, but also through external benchmarks and the needs and preferences of users and stakeholders (Detert et al., 2000).

We posit that an open climate is particularly conducive to implementing design thinking practices. Organisations that successfully apply design thinking practices interact with external sources, viewing such interactions not as threats or risks but as opportunities for enrichment and learning (Carlgrén & BenMahmoud-Jouini, 2022). In particular, the experimentation and inspiration phase of design thinking benefits from external inspiration to challenge internal ideas. Design thinking techniques such as user research, customer journey mapping, or co-creation rely on external opinions from users and stakeholders. Idea generation and problem-solving are also enhanced by involving and collaborating with external project partners (Carlgrén & BenMahmoud-Jouini, 2022). Thus, openness and appreciation of external ideas and partners align closely with the design thinking mindset, which is why we assume:

H2c: An open climate is conducive to using design thinking practices.

3.3. Team climates

While the organisational level provides the framework for implementing design thinking practices, the operational application usually occurs in project teams. Thus, we also consider specific team climates that might be conducive to design thinking practices. Based on relevant literature (Carlgrén & BenMahmoud-Jouini, 2022; Wrigley et al., 2020), we investigate the influence of shared leadership and team autonomy.

Shared leadership is “a dynamic, interactive influence process among individuals in groups for which the objective is to lead one another to the achievement of group or organisational goals or both” (Pearce & Conger, 2003, p. 1). In this approach, leadership responsibilities are distributed among group members rather than concentrated in a single supervisor, ensuring that the perspectives and contributions of all project members are valued and considered equally (Pearce & Conger, 2003). We suggest that shared leadership fosters the implementation of design thinking practices. In contrast to shared leadership, traditional leadership styles tend to prioritise individual work over group collaboration, often incorporating reward systems that focus on individual performance. This approach frequently views group work as inefficient and less promising (Carlgrén & BenMahmoud-Jouini, 2022; Detert et al., 2000). However, design thinking, as a collective problem-solving approach, underscores the importance of teamwork. Key practices such as brainstorming, problem-solving, and exploration are intended to be carried out collaboratively within a diverse team, where all opinions are regarded as equally valuable (Carlgrén, Elmquist, et al., 2016a; Liedtka, 2014). A supportive leadership style encourages this kind of collective teamwork. Teams that follow a shared leadership style may also see greater value in design thinking practices and, therefore, are more likely to adopt them. We therefore propose:

H3a: Shared leadership is conducive to using design thinking practices

Team autonomy is “the range of decisions over which a team has

discretion” and “the extent to which the team is responsible for its own work activities” (Gerwin & Moffat, 1997, p. 294). This allows project teams to work autonomously and make decisions without relying on other project teams or a central decision-making authority while still having the overall goals set by the project managers (Carlgrén & BenMahmoud-Jouini, 2022). Based on recent literature, we hypothesise a positive relationship between team autonomy and design thinking practices (Carlgrén & BenMahmoud-Jouini, 2022; Wrigley et al., 2020).

Giving project teams more autonomy demonstrates trust in team members' decision-making power. Consequently, results are more likely to be seen and valued as a team effort, not just a good leader's result. This makes project teams feel more valued and motivated to achieve better results. Furthermore, design thinking as an iterative work approach requires that teams independently decide when and how to switch between divergent and convergent thinking (Carlgrén & BenMahmoud-Jouini, 2022). Team members can best determine whether their ideas require testing or further brainstorming sessions (Wrigley et al., 2020). The idea of team autonomy allows and encourages this way of working and is therefore consistent with the design thinking mindset. Project teams that can make more autonomous decisions will likely see greater value in adopting design thinking practices. Therefore, we assume:

H3b: Team autonomy is conducive to using design thinking practices.

3.4. Moderating effects of project innovativeness

The impact of design thinking practices on project success and their susceptibility to different climates may vary depending on the specific project environment and context. Therefore, it is useful to consider the project conditions when analysing these relationships. Hereby, the degree of project innovativeness, ranging from incremental to radical, plays an important role (Garcia & Calantone, 2002). Various studies confirm that the efficiency of working practices depends on the degree of project innovativeness (Colombo et al., 2017; Ettlie et al., 1984; Salomo et al., 2007). In this study, we understand project innovativeness as the degree of technological novelty (Kock et al., 2011; Schultz et al., 2013). In a technologically novel environment, project teams can only partially rely on existing knowledge and techniques and must develop new ideas. Therefore, projects with higher innovativeness face higher task uncertainty, complexity, and ambiguity (Kock et al., 2011).

Scholars disagree on whether design thinking practices are more suited to radical or incremental innovation. On the one hand, they assume design thinking is only suited to pursue incremental innovation. The iterative cycle of user research, ideation, prototyping, and testing builds on and improves the ideas of the previous cycles so that it naturally leads to incremental improvements (Norman & Verganti, 2014; Verganti, 2011). Radical innovation, in contrast, would require completely new ideas. Norman and Verganti (2014) liken the human-centred iterative cycle to climbing a hill step by step, with the top representing the local maximum. The same applies to design thinking, where continuous improvement is achieved in each iterative loop, but the approach is trapped in a local maximum (Norman & Verganti, 2014). They argue that radical innovation would require moving to another hill to achieve a global maximum (Norman & Verganti, 2014).

On the other hand, scholars describe the potential of design thinking to pursue radical innovation. Brown (2008, p. 87) describes how “design thinkers pose questions and explore constraints in creative ways that proceed in entirely new directions”, and Gruber et al. (2015, p. 2) emphasize the capability of design thinking for breakthrough and disruptive innovations. Further, the literature claims that design thinking is well suited to solve “wicked problems” that involve large uncertainties and complexities (Buchanan, 1992; Fraser, 2007), which aligns with radical innovation's challenges. Liedtka (2015) argues that design thinking best suits uncertain and complex decision-making processes. Ben Mahmoud-Jouini et al. (2016) emphasise its efficiency for

exploratory projects with unclear goals and purposes. Due to this debate, we consider an empirical investigation of the moderator project innovativeness in the design thinking context useful.

We argue that positive moderation effects prevail regarding the relationship between design thinking practices and project success. As projects become more innovative, project teams face more complex problems, making design thinking practices more important. In the search for possible solutions, team members in more innovative and uncertain projects may be more willing to engage in new and innovative ways of working (Nguyen et al., 2018). With its iterative structure and creative “out of the box” thinking, design thinking is particularly suited to address such complex problems (Cooper et al., 2009; Dorst, 2011). It can help create a common working structure for project teams to better address these challenges to foster overall project success (Liedtka, 2020; Plattner et al., 2018). Therefore, we expect that as project innovativeness increases, the positive influence between design thinking practices and project success increases.

H4: Project innovativeness strengthens the relationship between design thinking practices and project success (positive moderation).

Regarding the influence of organisational and team climates on design thinking practices, we argue for positive moderation effects of project innovativeness. Higher project innovativeness brings greater uncertainties and challenges for organisations and project teams (Kock et al., 2011). In such environments, a supportive and innovative climate becomes increasingly critical. To address these challenges effectively, it is helpful for employees to experience psychological safety (Baer & Frese, 2003; Edmondson et al., 2004) and to show higher levels of commitment and motivation, which are fostered by a supportive climate. In addition, an innovative climate encourages unconventional and creative ideas, so employees are more likely to take risks and generate creative ideas without fear of negative consequences (Carlgrén & BenMahmoud-Jouini, 2022; Elsbach & Stigliani, 2018). This is particularly valuable for tackling complex problems in highly innovative projects. We, therefore, argue that as project innovativeness increases, so does the influence of a supportive and innovative climate on design thinking practices.

H5a: Project innovativeness strengthens the relationship between supportive climate and design thinking practices (positive moderation).

H5b: Project innovativeness strengthens the relationship between innovative climate and design thinking practices (positive moderation).

Furthermore, we argue for positive moderation effects of project innovativeness on the relationship between team climate and design thinking practices. Shared leadership and project team autonomy encourage collective teamwork and are therefore likely to be more important when the level of project innovativeness is higher. Shared leadership distributes the responsibility among team members, not just the project manager, and encourages each team member to give suggestions for improvement proactively (Pearce & Conger, 2003). This enhances the implementation of design thinking practice and fosters a positive, motivating, and constructive team atmosphere, which is more important in highly complex and innovative environments. When uncertainty is high, and project goals are unclear, teams need to be able to autonomously and spontaneously redefine and make decisions about the problem they are working on. This allows them to move flexibly between different design thinking phases and respond quickly to uncertain and innovative project environments without depending on other project teams and decisions (Carlgrén & BenMahmoud-Jouini, 2022). Therefore, we hypothesise that the positive effects of shared leadership and team autonomy on design thinking practices increase in more innovative and uncertain projects.

H5c: Project innovativeness strengthens the relationship between shared leadership and design thinking practices (positive moderation).

H5d: Project innovativeness strengthens the relationship between team autonomy and design thinking practices (positive moderation).

4. Method

4.1. Data and sample

We conducted a cross-sectional, multi-informant survey of mid-sized and large companies in various industries to explore the organisational factors influencing design thinking practices. The object of analysis is the project nested in a company (or a business unit in the case of large companies). The projects surveyed covered a wide range of topics. Mid-level managers, such as portfolio managers or project management office directors, were contacted because of their broad understanding of the project environment. These individuals were informed of the purpose and conditions of the survey and asked to distribute the survey to several project managers. We instructed project managers to rate their most recently completed project to minimise positive selection bias. As a result, our survey captured input from two types of respondents per company. This multi-informant approach mitigates the risk of common method bias (Podsakoff et al., 2003) and allows for separate analysis of company and project levels. The survey yielded completed responses from 379 projects nested in 93 companies/business units. Table 1 provides an overview of the sample.

4.2. Measurement

This study uses multi-item scales derived from the relevant literature. We sometimes slightly adapted the item wording to align with the study context. Informants responded to all items on a 7-point Likert scale ranging from 1 (“strongly disagree”) to 7 (“strongly agree”). We employed Confirmatory Factor Analysis (CFA) to assess construct validity of reflective constructs. We considered Cronbach’s alpha to assess scale reliability, with an acceptable cut-off value of 0.7 (Hair et al., 2010). We applied the cut-off criteria from Hu and Bentler (1998) to evaluate the CFA. The CFA ($\chi^2[df = 215] = 430.47$; RMSEA = 0.052; SRMR = 0.046; CFI = 0.93) had a good overall fit. The Appendix shows all item wordings.

Dependent variable. Project success was operationalised as a five-item formative scale with items based on previous literature (Hoegl & Gemuenden, 2001). They capture the achievement of planned profitability goals, the learning derived from the project, and the adherence to the project’s schedule, budget, and defined specifications. Project managers assessed this construct.

Mediator variable. We relied on construct measures from the literature to measure the application of design thinking practices in the projects (Loderer & Kock, 2021; Salzmann, 2020; Salzmann & Kock, 2018). Project managers evaluated ten items to assess the extent to which design thinking themes from Carlgrén et al. (2016) were present. These themes include user focus, problem framing, visualization, experimentation, and diversity. The collective weight of these five dimensions is indicative of the overall design thinking construct. Altering one of these dimensions would result in a concomitant alteration in the extent to which design thinking practices are employed, though not necessarily in any other dimensions (Jarvis et al., 2003). Additionally, these dimensions do not necessarily have to be correlated or possess analogous content (Jarvis et al., 2003). Consequently, design thinking practices constitute a formative rather than a reflective construct (Bollen, 2011; Jarvis et al., 2003). The items did not explicitly mention design thinking to circumvent potential halo bias.

Antecedents. The construct *supportive climate* ($\alpha = 0.80$) was operationalised with a five-item scale based on relevant literature (Eisenberger et al., 1986). This construct refers to the extent to which an organisation values the contributions of its employees and cares about their well-being. We measured this construct on the firm level using a weighted average of all project managers’ assessments. We operationalised the construct *innovative climate* based on established items (Kock et al., 2015; Kock & Georg Gemuenden, 2016) ($\alpha = 0.86$). This assessment entailed the aggregated evaluation of a five-item construct,

Table 1
Sample characteristics.

Industry		Revenue		Portfolio Budget		Employees	
Finance	24 %	<100 m EUR	16 %	<10 m EUR	24 %	<500	17 %
Engineering	18 %	100–500 m EUR	23 %	10–30 m EUR	30 %	500–2000	30 %
Electronics/IT	14 %	501–2000 m EUR	29 %	31–100 m EUR	21 %	>2000	53 %
Logistics/construction	14 %	>2000 m EUR	32 %	>100 m EUR	25 %		
Healthcare	7 %						
Chemicals/Pharma	6 %						
Utility/Public/Others	17 %						

Project Budget		Team size		Project Leader Tenure	
<200 K EUR	11 %	<3 team members	2 %	<2 years	6 %
200–500 K EUR	11 %	3–4 team members	6 %	2–3 years	17 %
501–2000 K EUR	20 %	>4 team members	94 %	>3 years	77 %
>2000 K EUR	58 %				

intending to gauge the organisation's inclination to foster and endorse creative and innovative ideas, as well as controversial opinions. We, therefore, aggregated the answers of all project managers and the middle manager to obtain a firm-level construct. The construct *open climate* ($\alpha = 0.81$) was drawn from relevant literature (Chesbrough, 2003; de Araújo Burcharth et al., 2014; Laursen & Salter, 2014). It captures an organisation's openness to collaborate with external project partners and willingness to look for external technologies and ideas. The middle management assessed this three-item construct. *Shared leadership* ($\alpha = 0.77$) is measured with a four-item scale based on previous literature (Hoch, 2013; Hoch et al., 2010; Pearce & Conger, 2003). Shared leadership within a project team places equal value on all opinions and promotes a collaborative environment where team members support and guide one another to enhance the team's collective performance. The project manager assessed this construct. Based on relevant literature (Gerwin & Moffat, 1997), *team autonomy* ($\alpha = 0.76$) was assessed on a three-item scale. This construct pertains to the capacity of the project team to exercise independent judgment regarding the project's scope and the coordination of tasks. The project manager assessed this three-item construct.

Moderator variable. *Project innovativeness* ($\alpha = 0.86$) measured the project's degree of technological innovativeness and novelty, with the construct items drawn from previous literature (Kock et al., 2011; Schultz et al., 2013). Project managers assessed the three-item scale.

Control variables. We included several control variables to mitigate potential confounding influences on project success and the adoption of design thinking practices. First, we controlled for *project team size*, which affects teamwork quality and project success (Hoegl, 2005). Smaller project teams might be better equipped to adopt and implement new work approaches, such as design thinking. Project team size was measured as the natural logarithm of the number of team members. Second, we included *project leader tenure* as a control variable, operationalized as the natural logarithm of years in the role. As less experienced project managers may encounter challenges in achieving project success (Savelsbergh et al., 2016), they may also have greater difficulties

in guiding their teams through the adoption of innovative work practices such as design thinking. Last, we controlled for *project budget*, measured in 1000 Euros and then transformed it into its natural logarithm. The availability of resources, including budget, can significantly influence project outcomes and the capacity to implement new methodologies. Table 2 shows the correlations and descriptives.

5. Results

This study applies random-effects regression with projects nested in companies to differentiate company- and project-level effects. Model 1 in Table 3 shows the relationship between design thinking practices and project success. Model 2 displays the effects of the antecedents, and models 3 and 4 display the moderation effects.

The control variables in Model 1 show that more experienced project leaders have, on average, higher success and that innovative projects are, on average, less successful. Model 1 further shows that design thinking positively relates to project success (unstandardized regression coefficient $b = .26$, $p = .000$). Thus, projects that apply design thinking practices are, on average, more successful. Overall, the model explains 15 % of the variance in project success, which is 8 % more than a model without design thinking practices (not shown in Table 3). The model explains not only variance within companies (12 %) but also between companies (33 %), which might indicate that some companies consistently achieve higher project success on average. Overall, these results show strong support for Hypothesis 1.

Regarding the antecedents in model 2, a supportive climate is conducive to using design thinking practices ($b = .20$, $p = .041$), which aligns with hypothesis 2a. Therefore, organisations with a supportive climate are more likely to adopt design thinking practices in their projects. However, hypothesis 2b, assuming a positive effect of an innovative climate, finds no support ($p = .060$). Our results find no evidence that organisations that foster an innovative climate are conducive to design thinking practices. Consistent with hypothesis 2c, an open climate favours using design thinking practices ($b = .12$, $p = .021$). This

Table 2
Correlations and descriptives.

Variables		M	SD	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1)	Project team size	2.68	0.83	1.00									
(2)	Project budget	8.73	3.46	0.16	1.00								
(3)	Project leader tenure	2.12	0.75	0.21	0.01	1.00							
(4)	Project success	5.40	0.97	0.12	−0.07	0.15	1.00						
(5)	Design thinking	4.60	1.06	0.09	0.00	0.02	0.25	1.00					
(6)	Project innovativeness	3.89	1.72	0.07	0.01	−0.01	−0.19	0.20	1.00				
(7)	Supportive climate	4.59	0.63	0.12	−0.07	−0.04	0.10	0.12	0.02	1.00			
(8)	Innovative climate	4.81	0.57	0.10	0.04	0.05	0.04	0.09	0.06	0.58	1.00		
(9)	Open climate	5.47	1.04	0.05	−0.06	0.11	0.15	0.13	−0.03	0.01	0.21	1.00	
(10)	Shared leadership	5.18	1.01	−0.05	−0.03	0.07	0.35	0.40	−0.02	0.09	0.19	−0.01	1.00
(11)	Team autonomy	4.72	1.27	0.07	−0.02	0.06	0.31	0.32	0.00	0.14	0.14	0.13	0.28

$N = 93$ companies/business units; $n = 379$ projects; M = mean; SD = standard deviation.

Table 3
Regression results.

	Model 1 (H1) Project Success	Model 2 (H2a-3b) Design Thinking	Model 3 (H4) Project Success	Model 4 (H5a-d) Design Thinking
Project team size (ln)	0.11 [0.08]	0.10 [0.07]	0.12 [0.08]	0.11 [0.07]
Project budget (ln)	-0.02* [0.01]	0.01 [0.01]	-0.02* [0.01]	0.00 [0.01]
Project leader tenure (ln)	0.15** [0.06]	-0.04 [0.10]	0.16** [0.06]	-0.02 [0.09]
Project innovativeness	-0.14** [0.03]	0.13** [0.03]	-0.14** [0.03]	0.14** [0.03]
Design thinking	0.26** [0.05]		0.26** [0.05]	
Supportive climate		0.20* [0.10]		0.19 [0.10]
Innovative climate		-0.21 [0.11]		-0.21 [0.11]
Open climate		0.12* [0.05]		0.12* [0.05]
Shared leadership		0.40** [0.06]		0.40** [0.06]
Team autonomy		0.17** [0.05]		0.17** [0.05]
Design thinking x project innovativeness			0.06* [0.03]	
Supportive climate x project innovativeness				0.11* [0.05]
Innovative climate x project innovativeness				-0.09 [0.06]
Shared leadership x project innovativeness				0.03 [0.03]
Team autonomy x project innovativeness				0.00 [0.03]
Constant	4.98** [0.22]	-0.24 [0.30]	4.93** [0.22]	-0.27 [0.29]
R2 (within)	0.12	0.30	0.12	0.31
R2 (between)	0.33	0.22	0.37	0.21
R2 (overall)	0.15	0.28	0.17	0.29

Random effects GLS regression; $N = 93$ companies/business units; $n = 379$ projects; * $p < 0.05$; ** $p < 0.01$; standard errors in brackets; interaction variables were mean-centred.

means that companies open to external ideas and opinions more likely support adopting design thinking practices in their projects.

In line with hypothesis 3a, shared leadership is a positive antecedent for design thinking practices ($b = .40, p = .000$), which shows us that organisations promoting a shared leadership style in their projects are more likely to adopt design thinking practices in their projects. Team autonomy also positively affects design thinking practices ($b = .17, p = .000$). Therefore, Hypothesis 3b is supported, and we can demonstrate that project teams with more autonomy are more likely to adopt design thinking practices.

Model 2 accounts for 28 % of the variance in design thinking practices, which is 23 % more than a model that includes only control variables (not shown in Table 3). This model explains a significant portion of within-company variance (30 %), indicating that there is also considerable variation between projects within the same company.

Regarding the interaction effects, model 3 shows a positive effect of project innovativeness on the relationship between design thinking

practices and project success ($b = .06, p = .037$). Thus, design thinking practices become more beneficial when project innovativeness is high. Compared to model 1, model 3 explains an additional 2 % of variance in project success. Overall, these findings support hypothesis 4.

To illustrate the nature of this interaction effect, Fig. 2 shows the simple slopes of the effect of design thinking practices on project success for high and low values of project innovativeness. A high score is the mean plus one standard deviation; a low score is the mean minus one standard deviation. It shows that using design thinking practices is associated with increased project success for low and high project innovativeness levels. However, the more innovative a project is, the stronger the effect of design thinking practices on project success.

Model 4 shows that the positive effect of a supportive climate on design thinking practices becomes stronger with increasing project innovativeness, as the interaction effect is positive ($b = .11, p = .034$). The model explains 1 % more variance in design thinking practices than model 2. Therefore, the data support hypothesis 5a. The assumed interaction effect between project innovativeness and innovative climate is not significant ($b = -0.09, p = .103$); thus, hypothesis 5b finds no support. Similarly, the interactions between project innovativeness and shared leadership ($b = .03, p = .288$) and team autonomy ($b = .00, p = .928$) are not significant. Thus, hypotheses 5c and 5d cannot be supported.

To explore the interaction effects of hypothesis 5a, Fig. 3 shows the corresponding simple slope analysis. A high value represents the mean plus one standard deviation, while a low value represents the mean minus one standard deviation. The results show that in highly innovative projects, a supportive climate positively influences the implementation of design thinking practices. However, when project innovativeness is low, there is no significant relationship between a supportive climate and the use of design thinking practices. Therefore, especially in innovative project settings, fostering a supportive climate facilitates the use of design thinking practices.

6. Discussion

This study set out to provide empirical evidence that design thinking serves as a valuable work approach for achieving project success, particularly in complex, innovative, and exploratory projects where traditional, more rigid project management methods often reach their limitations (Atkinson et al., 2006; BenMahmoud-Jouini & Carlgren, 2023; Lenfle, 2008). Our findings demonstrate that design thinking may complement classical project management approaches, especially in highly innovative contexts where conventional methods lack flexibility and openness. This finding enriches the project management literature by highlighting design thinking as a robust approach for navigating the challenges of complex, innovation-driven projects (Atkinson et al., 2006; Lenfle, 2008). Additionally, we aimed to address how organisational and team climates affect the adoption of design thinking practices. Our results reveal that successfully integrating design thinking practices depends on the prevailing organisational and team climates and how such climates fit with the values of design thinking. These findings contribute empirical validation to the work of BenMahmoud-Jouini and Carlgren (2023) by demonstrating the critical role of project conditions in fostering the adoption of design thinking.

The results demonstrate that organisations with a supportive climate are more likely adopt design thinking practices. As employees feel more valued and motivated to achieve organisational goals, project team members may be more motivated to engage in a new way of working, such as design thinking. Therefore, a supportive climate is conducive to applying design thinking practices. Relevant literature suggests that a supportive climate fosters a sense of meaning and appreciation among employees, which in turn enhances their intrinsic motivation and commitment (Eisenberger et al., 1986; Rhoades & Eisenberger, 2002). Such a climate appears particularly conducive to the adoption of new work approaches, like design thinking, within project environments.

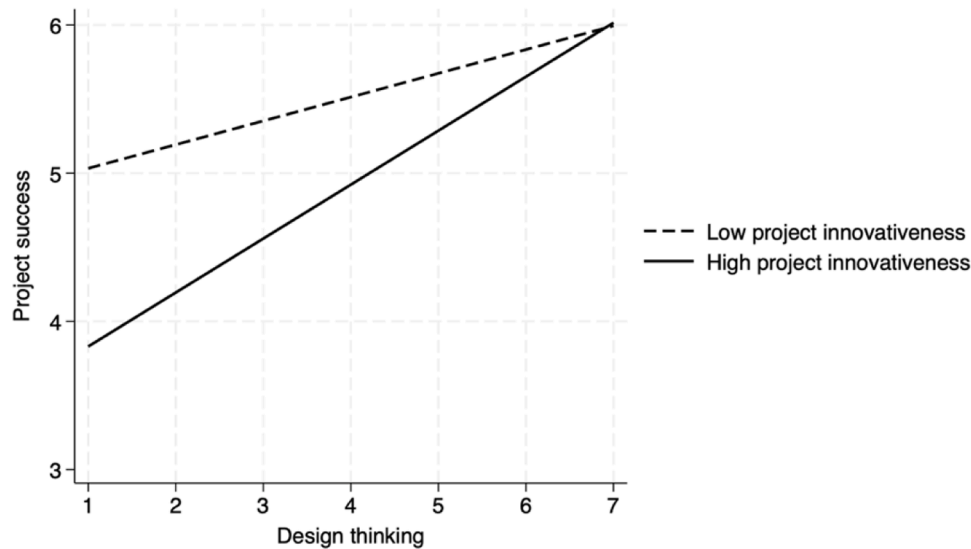


Fig. 2. Simple slopes analysis for the effect of design thinking on project success (moderated by the degree of project innovativeness).

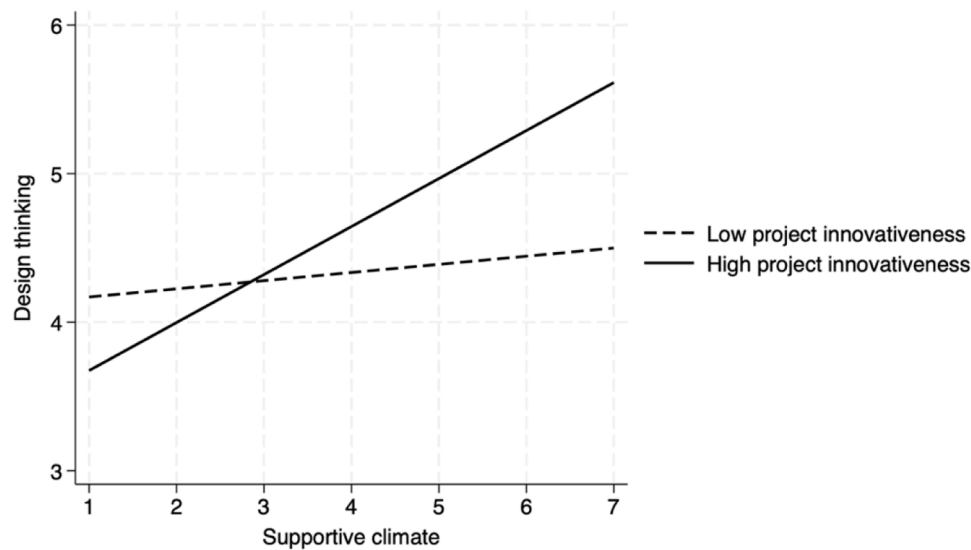


Fig. 3. Simple slopes analysis for the effect of a supportive climate on design thinking (moderated by project innovativeness).

Surprisingly, the results do not support the assumption that organisations with an innovative climate are more likely to adopt design thinking practices. However, this non-significant effect does not necessarily indicate that the effect does not exist; rather, we were unable to demonstrate it, possibly due to a Type II error. One potential reason for this could be the study's sample size, which may have been insufficient to detect the effect. More likely, shared variance might have contributed to the non-significant result. Given that we tested all antecedents with a single model and some of them exhibited strong correlations (e.g., innovative climate and supportive climate, $r = 0.58$), it is possible that the effect of innovative climate on design thinking practices became non-significant because of this shared variance. Due to the overlap in variance explained by an innovative climate and a supportive climate, the unique contribution of an innovative climate could not be identified. Further research is needed to clarify this relationship.

Design thinking is a way of working that relies heavily on external sources of inspiration, a notion highlighted in the qualitative study by BenMahmoud-Jouini and Carlgren (2023), for which our study provides quantitative support. An openness to external sources encourages the involvement of users and stakeholders in the development process.

Organisations open to external ideas and project partners are, therefore, more likely to see the benefits of design thinking practices and more willing to engage in this way of working. Thus, an open climate is conducive to applying design thinking practices.

In addition, project teams that promote a shared leadership style are more likely to engage in design thinking practices. Shared leadership promotes the idea that each project member's opinion counts equally and that all team members work together to achieve the overall goal (Pearce & Conger, 2003). As suggested by Carlgren, Elmquist, et al. (2016a) and Liedtka (2014), key practices of design thinking—such as brainstorming, problem-solving, and exploration—are inherently collaborative and thrive within diverse teams where all perspectives are valued equally. Building on this, our findings demonstrate that shared leadership aligns closely with this collaborative ethos, thereby increasing the likelihood of adopting design thinking practices. Furthermore, project teams with high autonomy, with minimal reliance on other teams or a central leader, are more prone to adopt design thinking practices. Accordingly, this study provides empirical evidence that substantiates the anecdotal findings by Carlgren and Ben Mahmoud-Jouini (2022) and Wrigley et al. (2020). As an innovative and

iterative approach, design thinking entails teams navigating between divergent and convergent thinking. Autonomous project teams inherently facilitate this independent workflow and decision-making process, rendering team autonomy a positive enabler of design thinking.

Collectively, these findings suggest that organisations are more likely to adopt design thinking practices in their projects when they foster a supportive and open climate while also enabling project teams to work autonomously and embrace shared leadership. All four constructs promote open-mindedness and appreciation of internal and external opinions. They lead to higher employee commitment and foster collective, autonomous teamwork and a positive work environment internally and with external partners. As introducing a new work approach, such as design thinking, is often fraught with high uncertainty and complexity, a positive and tolerant attitude among team members is even more important for successful implementation. It keeps employees motivated and more willing to engage in design thinking without fear of unfamiliar working methods and possible failure.

Our results also demonstrate that applying design thinking practices in projects is associated with higher project success, which aligns with previous studies in innovation management (Roth et al., 2020; Seidel & Fixson, 2013). The iterative, user-centred work approach fosters creative, open-minded thinking and encourages companies to regularly challenge their ideas to achieve better project performance (Brown & Katz, 2011; Liedtka, 2018; Roth et al., 2020; Seidel & Fixson, 2013). Regarding the moderation effects of project innovativeness, our results show that design thinking practices become more important in highly innovative and complex projects. Apparently, design thinking is particularly suited to address complex problems in highly innovative and uncertain project environments (Cooper et al., 2009; Dorst, 2011). This finding supports the view in project management literature that particularly complex projects can benefit from design thinking practices (BenMahmoud-Jouini & Carlgren, 2023; Liedtka & Locatelli, 2023). Further, a supportive climate is even more important for successfully applying design thinking practices when project innovativeness is high. Higher project innovativeness brings greater uncertainties and challenges for organisations and project teams. To better face these challenges, it is helpful if employees feel psychologically safe (Baer & Frese, 2003; Edmondson et al., 2004) and show higher levels of commitment, which is encouraged by a supportive climate. Contrary to our expectations, our findings did not demonstrate positive moderation of project innovativeness on the associations of innovation climate, shared leadership, and team autonomy with design thinking practices. This suggests that irrespective of the project's innovativeness, shared leadership and team autonomy provide equivalent support for implementing design thinking practices.

In summary, this study bridges the literature on design thinking's impact on project success with research on how project conditions influence its adoption, offering quantitative evidence to support and extend existing qualitative insights (BenMahmoud-Jouini & Carlgren, 2023; Elsbach & Stigliani, 2018; Wrigley et al., 2020).

6.1. Implications for theory

Our findings provide several contributions to project management and design thinking theory. First, this study contributes to the literature by demonstrating that design thinking serves as a valuable alternative to traditional project management approaches in enhancing project success. This contributes to the extant body of research that critiques the limitations of classical project management methods in the context of innovative, uncertain, and exploratory projects (Atkinson et al., 2006; BenMahmoud-Jouini & Carlgren, 2023; Lenfle, 2008). Lenfle (2008)

highlights the missing link between project management and innovation and how classical project management approaches, characterised by rigid planning and predefined objectives, are at odds with the idea of creating innovation.

Second, the results of this study emphasise the value of design thinking in projects characterised by high levels of innovativeness and uncertainty. We show that the design thinking practices, with their flexible, user-centred, and open way of thinking and working, are an ideal work approach to provide appropriate methods for projects to foster innovation and exploration. As we empirically demonstrate, the relationship between design thinking and project success becomes stronger in highly innovative, novel projects. Thus, we enhance the existing literature on managing uncertainty and risk in project management (Atkinson et al., 2006). By offering evidence that design thinking is well-suited to navigate ambiguous and uncertain projects, we underline its value in addressing these complex challenges. Previous conceptual findings on the role of design thinking in project management argue that design thinking practices can promote exploration, stakeholder involvement, and strategising in project management and, thus, help deal with exploratory and uncertain projects (BenMahmoud-Jouini & Carlgren, 2023). We support these findings and provide quantitative evidence on how design thinking practices are particularly suited to address complex, uncertain, and innovative issues. With this, we also enrich the debate on whether design thinking practices are more useful for projects pursuing radical or incremental innovation (Micheli et al., 2019; Norman & Verganti, 2014). Our analysis shows that design thinking practices are more helpful in innovative and uncertainty-prone projects facing similar challenges to radical innovations.

Third, we contribute to the extant body of research by demonstrating how team and organisational climates characterised by support, openness, and autonomy can foster the adoption of design thinking in projects. Existing literature has highlighted formal project management structures and processes as potential barriers to design thinking practices (Hölzle & Rhinow, 2019; Mahmoud-Jouini et al., 2016). The present study extends this by investigating how organisational and team climates influence the adoption of design thinking practices, thereby expanding the scope beyond the impact of formal project management processes. We provide novel empirical insights into how upstream organisational climate conditions can support the implementation of design thinking practices and help mitigate some project management barriers identified in previous research. We thus show that, beyond the structural and strategic conditions identified in prior research, different climates greatly influence whether project teams adopt design thinking practices. Our study enriches the literature on design thinking and organisational climate. In previous studies, design thinking was often understood as a set of tools and techniques neglecting the associated mindset (Liedtka, 2018; Micheli et al., 2019; Seidel & Fixson, 2013). Studies analysing the interdependence of design thinking and the organisational environment are conceptual (Elsbach & Stigliani, 2018; Wrigley et al., 2020) or qualitative (Carlgren & BenMahmoud-Jouini, 2022). They highlight the important role of the interplay between a specific organisational mindset and design thinking practices (Carlgren & BenMahmoud-Jouini, 2022; Elsbach & Stigliani, 2018; Wrigley et al., 2020). Drawing on these findings, we show that an organisational climate characterised by open, autonomous, collaborative, and supportive values is conducive to implementing design thinking practices. This study extends previous findings with quantitative evidence and further demonstrates the moderating role of project innovativeness. Our findings highlight the importance of fit between design thinking practices and the adopting organisational climate. This also adds to the

literature on the role of cultural fit between a new work approach and the adopting organisation's culture (Ansari et al., 2010; Canato et al., 2013). To the best of our knowledge, this is the first study to provide large-scale quantitative evidence on organisational antecedents for design thinking practices, responding to the call for contributions of Elsbach and Stigliani (2018).

Finally, this study enriches the innovation management literature by underscoring the role of fostering project conditions in enabling the implementation of design thinking, with implications that may also be relevant for other innovative work approaches. As design thinking can be understood as a management innovation (Birkinshaw et al., 2008; Carlgren & BenMahmoud-Jouini, 2022), it encounters obstacles that partly correspond to innovation barriers. Similar to our findings, previous literature highlights the interplay between culture and management innovation (Klein & Sorra, 1996; Waarts & Van Everdingen, 2005). The antecedents we examined are consistent with an organisational climate conducive to innovation (Carlgren & BenMahmoud-Jouini, 2022; Detert et al., 2000), such as open innovation (Chesbrough, 2003), a collaborative, shared leadership style that involves all team members (Hoch, 2013), a supportive climate (Le & Lei, 2019), as well as autonomously working project teams (Carlgren & BenMahmoud-Jouini, 2022). Thus, our findings may not only account for organisations engaging in design thinking practices but also for other management innovations, such as lean (Dowson et al., 2024) or agile development (Meier & Kock, 2023).

6.2. Implications for practice

Our study provides important insights for practitioners. We provide evidence from different industries that the application of design thinking practices promotes project success. The more innovative a project is, the stronger this relationship becomes, so particularly innovative projects should consider integrating design thinking practices. Further, organisations benefit from our findings in assessing their thinking and working with the design thinking mindset. The results show that a supportive, collaborative, and open climate is conducive to implementing design thinking practices. Although changing organisational climates is difficult, our findings show managers and project leaders the importance of adapting their organisational climate to design thinking. At the team level, project leaders can proactively cultivate a shared leadership style and promote autonomous work within their project teams. At the firm level, efforts should focus on fostering a supportive and open work environment. This can increase organisational and project team engagement and open-mindedness, creating a more open climate for design thinking practices. Overall, we create awareness of how organisations are positioned to be more likely to apply design thinking practices and ultimately increase their overall project success. In addition, organisations can use these insights to design targeted training programs that build design thinking skills across teams. By equipping employees with empathy, creativity, and iterative problem-solving skills, organisations can build a better-prepared workforce to tackle complex and ambiguous challenges. These training initiatives should focus not only on technical tools but also on fostering collaboration and psychological safety within teams. Overall, we raise awareness of how organisations can position themselves to be more likely to apply design thinking practices and ultimately increase their overall project success.

6.3. Limitations and future research

This study is not without limitations. Our analysis relies on cross-sectional survey data, which is correlational in nature and limits the ability to draw causal conclusions. Although we used multiple informants at the project and company levels for 379 projects nested in 93 companies to mitigate endogeneity concerns, the possibility of common method bias cannot be ruled out (Podsakoff et al., 2003).

Future research could benefit from longitudinal designs to better understand the causal dynamics between organisational and team climates and the adoption of design thinking practices. For instance, such studies could examine how climates evolve as employees gain experience with the work approach or how design thinking practices shape workplace climates. As Chen and Venkatesh (2013) suggest, design thinking practices may embed new mindsets, beliefs, and routines into organisations over the long run, underscoring the importance of studying these bidirectional effects.

Another avenue for future research is to explore the role of openness at the team level, as this study focused on the organisational level. Examining openness at the team level could provide additional insights into its role in driving innovation and collaboration within organisations. Similarly, a more comprehensive assessment of shared leadership and team autonomy—drawing on input from multiple team members rather than only team leaders—could help reduce single-source bias and provide a more nuanced understanding of these constructs.

While we examined the moderating role of project innovativeness, future studies could explore additional interactions, such as the level of expertise in applying design thinking practices. This could shed light on how experience influences the adoption and impact of design thinking practices. Future research could also further explore the type of projects for which design thinking is particularly useful. Prior research suggests its relevance in open innovation contexts (Loderer & Kock, 2021), but future work could investigate specific types of collaborations, such as startup-industry or university-industry collaborations. Beyond open innovation projects, identifying the specific project contexts where design thinking practices increase project success remains an important research goal.

Finally, in light of the increasing shift to remote and hybrid work, it could also be valuable to explore the role of digital tools and technologies in facilitating or hindering the adoption of design thinking practices. For example, studies could examine the impact of collaborative software, virtual reality tools, or AI-driven ideation support systems on team creativity and engagement in design thinking projects. This could reveal how technological enablers interact with different organisational and team climates.

CRediT authorship contribution statement

Maria Loderer: Writing – original draft, Methodology, Data curation, Conceptualization. **Alexander Kock:** Writing – review & editing, Validation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors have no conflict of interest to declare.

Appendix: Construct measurement

Construct	Items	Lambda
Project Success	The project achieved a high schedule adherence.	
<i>Project Manager</i>	The project achieved a high budget adherence.	
Formative construct	The project fulfilled the defined specifications.	
	The product/project result achieved the planned profitability goals (e.g., ROI).	
	During the course of the project, we gained valuable insights.	
Design Thinking	<i>User focus:</i>	
<i>Project Manager</i>	We focused on the experiences of users and let this understanding guide our work.	
Formative construct	We sought to understand users' unarticulated needs and pain points.	
	<i>Diversity:</i>	
	We sought diverse perspectives and inspirations.	
	We were very interdisciplinary, involving technical, design, and business disciplines at most times.	
	<i>Problem framing:</i>	
	We focused not only on the specific issue under consideration, but also incorporated a holistic perspective on its environment.	
	We challenged and reframed the initial problem to expand both the problem and the solution space.	
	<i>Visualisation:</i>	
	We created rough representations/visualisations of the product or service as soon as possible.	
	We regularly presented interim results of our project (e.g., in the form of prototypes or minimum viable products).	
	<i>Experimentation:</i>	
	We operated highly iteratively, moving back and forth through ideation, concept generation, prototyping, and testing.	
	The goals were continuously adjusted during the course of the project based on gained knowledge.	
Supportive Climate	The company values my contribution to its success.	0.69
<i>Average of Project Managers</i>	The company does not recognise any additional efforts on my part. (inverted)	0.32
$\alpha = 0.80$	The company wants to offer me the best possible project for which I am qualified.	0.64
	The company tries to make my job as a project manager as interesting as possible.	0.68
	The organisation cares about my general satisfaction.	0.72
Innovation Climate	In our organisation...	
<i>Average of Project & Middle Managers</i>	...employees can express their thoughts and true feelings regarding their job.	0.77
$\alpha = 0.86$	...nobody will pick on employees even if they have different opinions.	0.72
	...employees are given sufficient responsibility, resources, and freedom to work independently.	0.71
	...we emphasise creativity and innovativeness.	0.60
	...supervisors encourage unconventional ideas.	0.59
Open Climate	Please evaluate below the cooperation with external partners in your organisation/ business unit.	
<i>Middle Manager</i>	We are open to cooperation with external project partners.	0.61
$\alpha = 0.81$	We also search outside our organisation's boundaries for project partners with innovative ideas.	0.81
	We are open to the application of ideas and technologies from outside the company.	0.91
Shared Leadership	Each team member helped to frame the vision for this team.	0.59
	Team members collaborated with one another in making decisions that affected this team.	0.79
<i>Project Manager</i>		
$\alpha = 0.77$	The opinion of each member counted when they shared their perceptions regarding a situation facing the team.	0.74
	All team members proactively made constructive suggestions for improving how things operate within the team.	0.69
Team Autonomy	The project had a high degree of autonomy.	0.70
<i>Project Manager</i>	The project was allowed to autonomously adapt its scope to current customer requirements.	0.79
$\alpha = 0.76$	The project was allowed to autonomously coordinate tasks with other projects.	0.64
Project Innovativeness	At the beginning of the project, we did not have the necessary technical knowledge.	0.79
<i>Project Manager</i>	At the beginning of the project, we had little practical experience in the application of the required technology.	0.93
$\alpha = 0.86$	In our project, we could only partially rely on the existing technological competence of the company.	0.74

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